Per-Slice Multivariate Refinement of g_m

Notes

1 Per-slice refinement

Following the multivariate Borodin identity (thesis, Section 4.1), we refine the generating function g_m by assigning a separate variable to each partition in the CPP triple. For a rank-3 cylindric plane partition $\nu = (\nu_0, \nu_1, \nu_2)$ with profile $c = (c_0, c_1, c_2)$, define the *per-slice multivariate generating function*:

$$g_m(z_0, z_1, z_2) \;=\; \sum_{\substack{\nu \in \mathrm{CPP}(c) \\ \max(\nu) = m}} z_0^{|\nu_0|}\, z_1^{|\nu_1|}\, z_2^{|\nu_2|}.$$

The univariate specialization $z_0 = z_1 = z_2 = q$ recovers the standard $g_m(q) = [z^m]\,\mathrm{GK}_c(z, q)$.

In the thesis, this refinement arises from replacing $z^{|\mathfrak{c}|}$ by $z_0^{|\mu_0|} z_1^{|\mu_1|} \cdots z_{T-1}^{|\mu_{T-1}|}$ on the left-hand side of the Borodin identity, with corresponding substitutions on the product side:

$$\begin{aligned} z^{nT} &\mapsto z_0^n z_1^n \cdots z_{T-1}^n, \\ z^{j-i+nT} &\mapsto z_0^n \cdots z_i^n\, z_{i+1}^{n+1} \cdots z_j^{n+1}\, z_{j+1}^n \cdots z_{T-1}^n \quad (i < j). \end{aligned}$$

2 Computational observations

`multivariate_Q.py` computes $g_m(z_0, z_1, z_2)$ for various profiles. Key findings:

1. **Specialization verified.** For all tested profiles ($d = 1, 2, 4, 5$) and $m \le 3$, the specialization $z_i = q$ matches the univariate $g_m(q)$ exactly.

2. **All coefficients of g_1 equal 1.** For $m = 1$, every CPP has a distinct slice-weight triple $(|\nu_0|, |\nu_1|, |\nu_2|)$, so all coefficients of $g_1(z_0, z_1, z_2)$ are 1. This holds because $\max(\nu) = 1$ forces each $\nu_k = (1^{|\nu_k|})$, so the partition is determined by its size.

3. **No S_3 symmetry.** The polynomial g_m is not symmetric under permutation of z_0, z_1, z_2, reflecting the asymmetry of the profile $c = (c_0, c_1, c_2)$.

4. **Cyclic staircase for $d = 1$.** For $c = (1, 0, 0)$ and $m = 1$, the slice weights advance as

   $$(1,0,0),\ (1,1,0),\ (1,1,1),\ (2,1,1),\ (2,2,1),\ (2,2,2),\ (3,2,2),\ \ldots$$

   cycling through which slice increases by 1 at each step.

5. **Collisions at $m \geq 2$.** For $m \geq 2$, distinct CPPs can share the same slice-weight triple (e.g. partitions (2) and (1, 1) both have size 2), so coefficients exceed 1.

Remark 1. *This is distinct from the* per-ring *refinement $t_1^{b_1} \cdots t_m^{b_m}$ where b_k is the number of cells at height k. The per-ring version does not specialise correctly to $(q)_m \cdot g_m$ because $\prod_{k=1}^{m}(1-t_k)$ specialises under $t_k = q$ to $(1-q)^m$, not $\prod_{k=1}^{m}(1-q^k) = (q)_m$.*